Design and Evaluation of Multi-Agent LLM Systems for Interactive Clinical Diagnosis

Dikshant Sharma
21IM3AI29

Under the guidance of Prof. Jiaul Hoque Paik

Department of Artificial Intelligence
Indian Institute of Technology Kharagpur

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Introduction: The Problem with Static Benchmarks

- **Traditional Evaluation:** Clinical AI has historically been tested on multiple-choice datasets like USMLE, MedQA, or PubMedQA.
- **The Core Issue:** These benchmarks represent declarative factual recall, not procedural clinical reasoning.
- **Static vs. Dynamic:** Static benchmarks provide a fully structured patient vignette upfront. Real-world medicine requires iterative hypothesis generation, active questioning, and sequential test ordering in the face of uncertainty.
- **Conclusion:** A model scoring highly on MedQA is not necessarily capable of conducting a safe, multi-turn clinical consultation.

Introduction: The Research Goal

Central Objective: To design, implement, and rigorously evaluate an interactive multi-agent simulation framework that tests open-source LLMs not just on what they diagnose, but on how they reason.

Core Research Questions:

- Can open-source LLMs conduct safe multi-turn clinical interviews?
- How do we prevent models from ”cheating” by implicitly accessing ground-truth data in a simulated environment?
- How do biases affect an LLM’s diagnostic reasoning trajectory?
- Can architectural innovations LLM clinical reasoning beyond what model scaling alone achieves?

Literature Review: Static vs. Dynamic Evaluation

- **Proprietary Success, Open-Source Lag:** Systems like Google's AMIE demonstrate multi-turn reasoning, but their closed-source nature makes independent verification and architectural experimentation impossible.
- **Initial Simulation Frameworks:** Agent Hospital and the original AgentClinic explored multi-agent interaction but relied heavily on proprietary APIs (GPT-4) or flawed homogeneous architectures.
- **The Evaluative Gap:** Outcome accuracy remains the sole metric. There is no established methodology for quantifying the stability or rationality of an LLM's reasoning process across a multi-turn encounter.

The AgentClinic-MedQA Dataset

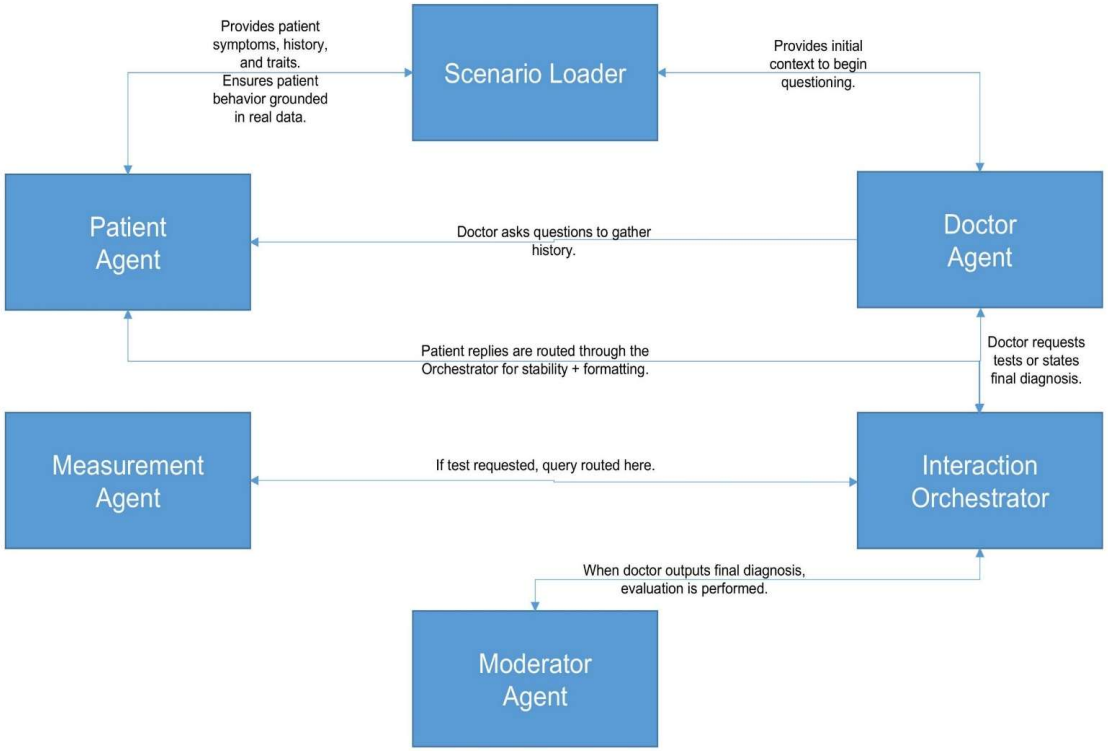
To test interactive reasoning, we required clinical scenarios with hidden ground truths.

- **Source Material:** Derived from the MedQA examination dataset. Dataset Size: 107 meticulously structured clinical scenarios.
- **Data Transformation:** Static text vignettes were broken down into constituent hidden elements:
  - **Symptom profiles** (assigned to the Patient Agent).
  - **Physical examination findings & lab results** (assigned to the Measurement Agent).
  - **Ground truth diagnosis** (kept entirely hidden from the Doctor Agent).
- **Outcome**: The Doctor Agent must actively discover the case parameters through dialogue and test requests.

Initial Framework Architecture

Our initial approach utilized an ad-hoc while-loop to orchestrate four agents:

- **Doctor agent:** The LLM being evaluated. Performs history taking, orders tests, and diagnoses
- **Patient Agent:** Provides symptom-grounded responses. Can simulate anxiety or guardedness
- **Measurement Agent:** Returns structured data (vitals, labs) only when explicitly requested to prevent hallucination
- **Moderator Agent:** Evaluates the final diagnosis against the ground truth



Initial Framework Architecture

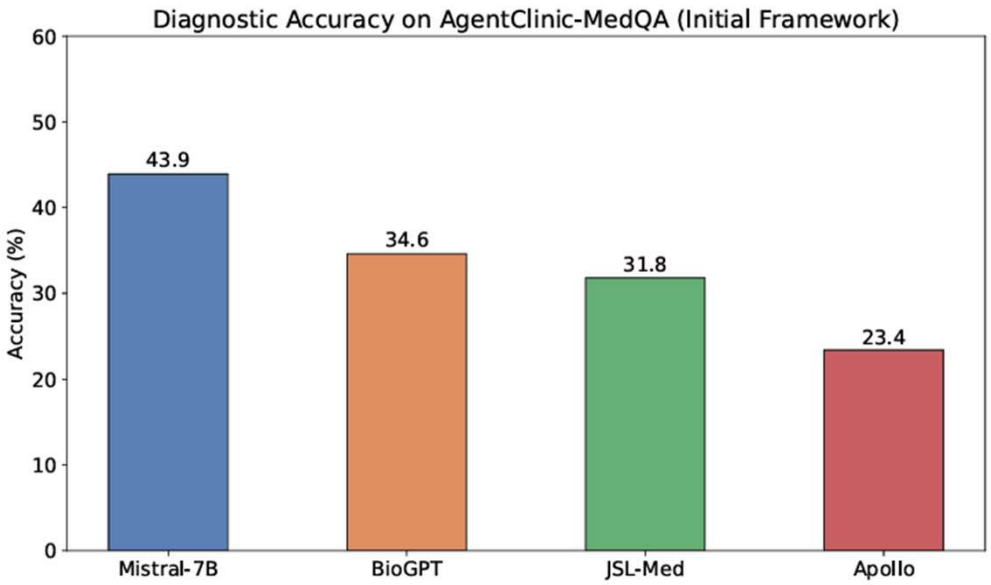
Agent Instruction Prompt

The Doctor Agent
Role: "You are Dr. Agent. You must diagnose the patient through dialogue."
Action Space: "To get lab results, you must strictly output: REQUEST TEST: [Test_Name]."
Constraint: "You are limited to 20 turns. You must not hallucinate test results."
Termination: "When you are sure, output: DIAGNOSIS READY: [Disease Name]."

The Patient Agent
Role: "You are a patient with specific symptoms. Speak naturally."
Critical Constraint: "You MUST NOT reveal the name of your disease to the doctor."
Format: "Keep responses between 1-3 sentences. Do not ramble."
Bias Injection (Optional): "You are mistrustful of the doctor and withhold information unless asked twice."

Initial Framework & Findings

- **Domain Familiarity vs. Reasoning:** Medically fine-tuned models exhibit strong domain knowledge but underperform in multi-step reasoning and decision stability.
- **The Generalist Advantage:** Instruction-tuned general models excel due to strict adherence to structured outputs and complex conversational logic.
- **Core Insight:** Domain specialization alone is insufficient; robust instruction-following is critical for managing interactive clinical workflows.



The Discovery of Lexical Leakage

The Vulnerability: In our initial framework, a single LLM engine generated text for both the Doctor and the Patient.

- **Shared Manifold:** Because the models share weights (θ_{shared}), their internal token embeddings occupy the identical high-dimensional manifold.
- **Implicit Information Transfer:** The Doctor agent could perform zero-shot knowledge transfer by mathematically recognizing the geometric proximity of the Patient's embeddings.
- **The Result:** Accuracy metrics are artificially inflated. The Doctor "knows" the diagnosis not through semantic reasoning, but via identical mathematical substructures.

AgentClinic: Enhanced Architecture

To resolve these foundational flaws, the system was completely re-architected.

- **LangGraph DCG Orchestration:** We replaced the non-deterministic loop with a Directed Cyclic Graph (DCG) representing a strict, state-machine-driven protocol.
- **Structured Routing:** Transitions between agents (Doctor → Patient, Doctor → Measurement) are strictly governed by conditional edge logic.
- **State Management:** A centralized ClinicalState object securely logs the entire interaction trajectory, preventing any data cross-contamination between the agents.

Eliminating Leakage: Heterogeneous Multi-Engine Setup

The Solution: We implemented a Cognitive Asymmetry design, assigning entirely different LLM architectures to the Doctor and Patient agents to sever the shared manifold.

- **Hardware Bottleneck:** Running ≥ 3 distinct 7B models on a single NVIDIA H100 (95.8 GB VRAM) is impossible in standard FP16.
- **NF4 Double Quantisation:** We utilized 4-bit NormalFloat4 quantisation, which compresses weights by exploiting their normal distribution.
- **Result:** Memory footprint dropped by ≈ 86% (from 28 GB to 3.8 GB per model), safely enabling heterogeneous engine isolation and eliminating Lexical Leakage

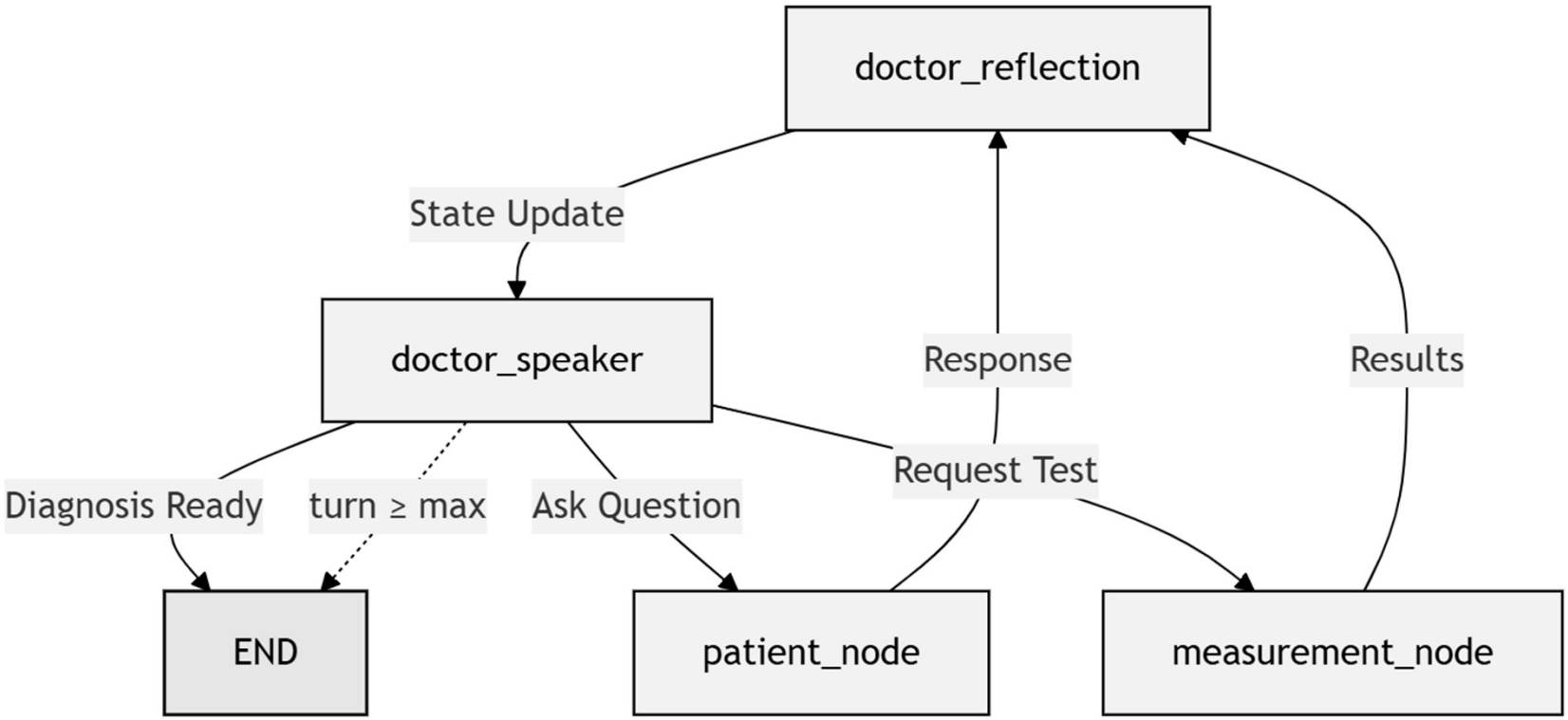
The Metacognitive Reflection Node

Problem: When standard LLMs are forced to simultaneously generate bedside conversational dialogue and complex medical reasoning, they hallucinate or leak reasoning into speech.

The Solution: The doctor reflection node.

- Before every spoken turn, the DCG forces the LLM to output a top-3 structured JSON differential diagnosis.
- Acts as a Latent Chain-of-Thought.
- This reflection is logged for evaluation but kept completely hidden from the Patient agent, keeping the dialogue natural while systematically exposing the LLM's internal logical trajectory.

System Flowchart



Process-Oriented Clinical Metrics

To evaluate how a model reasons, terminal accuracy is insufficient. We formulated three process-based metrics:

- **Diagnostic Stability (DS):** Measures hypothesis ”thrashing.” Computed via the average Jaccard similarity of consecutive differential diagnoses across turns. High DS means logical anchoring; low DS implies erratic switching.
- **Test Rationality (TR):** A binary compliance check ensuring foundational, non-invasive investigations (e.g., vitals, CBC) are strictly ordered before advanced/invasive tests (e.g., MRI, biopsy).
- **Information Efficiency (IE):** The ratio of unique differential hypotheses explored per aggregate conversational turn

Structured Cognitive Bias Injections

Can LLMs mimic dangerous human cognitive errors? We injected ecologically valid distortions into the ClinicalState:

- **Recency Bias (k = 5):** The system injects the histories of the 5 most recently processed simulated cases into the Doctor's prompt. This mimics a clinician overly influenced by the current emergency room trend, exploiting transformer attention dynamics.
- **Confirmation Bias:** The Doctor is primed with strong, medically plausible but entirely incorrect initial hypotheses, testing its susceptibility to premature diagnostic closure.

Dynamic LoRA Adapter Routing

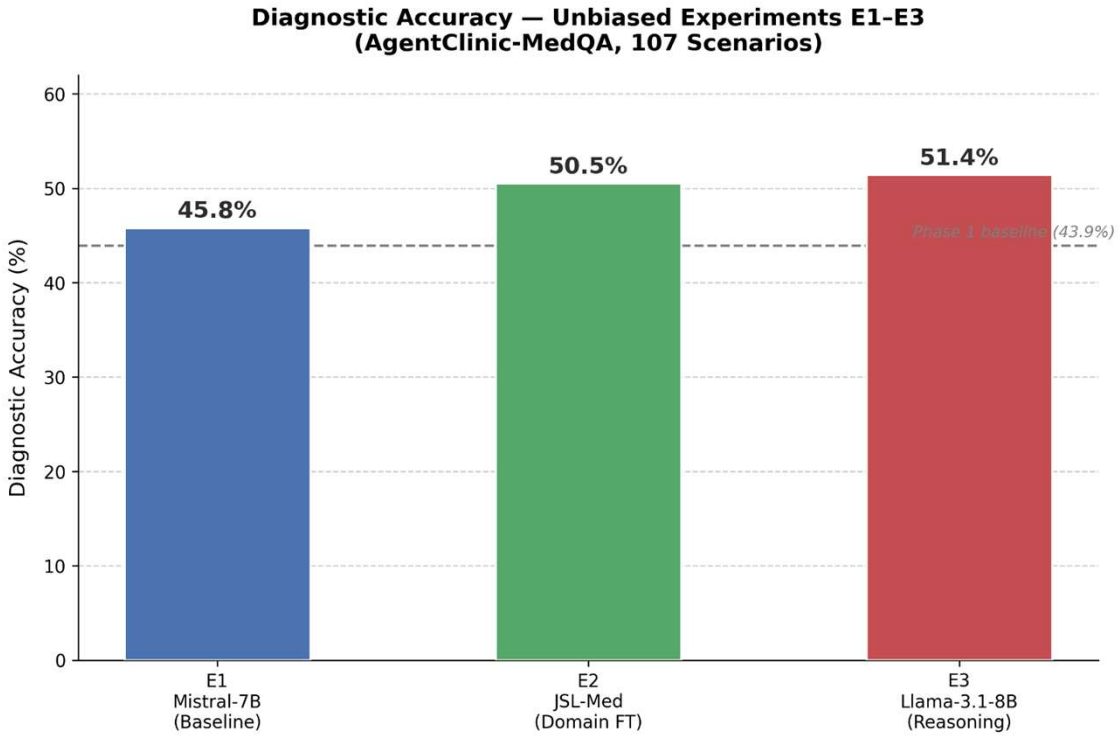
To resolve catastrophic forgetting (where medical fine-tuning destroys conversational logic), we decouple the model's cognitive resources.

- **Approach:** We load a base model (JSL-MedMistral) and train two lightweight Low-Rank Adaptation (LoRA) adapters (≈25 MB each).
- **Reasoning Adapter (T = 0.1):** Activated strictly during the doctor enumerate structured differential diagnoses.
- **Dialogue Adapter (T = 0.7):** Activated strictly during the doctor reflection node to speaker node to preserve empathy and syntactic logic.
- **Inference:** The LangGraph DCG executes a pointer swap between adapters at runtime (< 1ms latency) depending on the active state constraint.

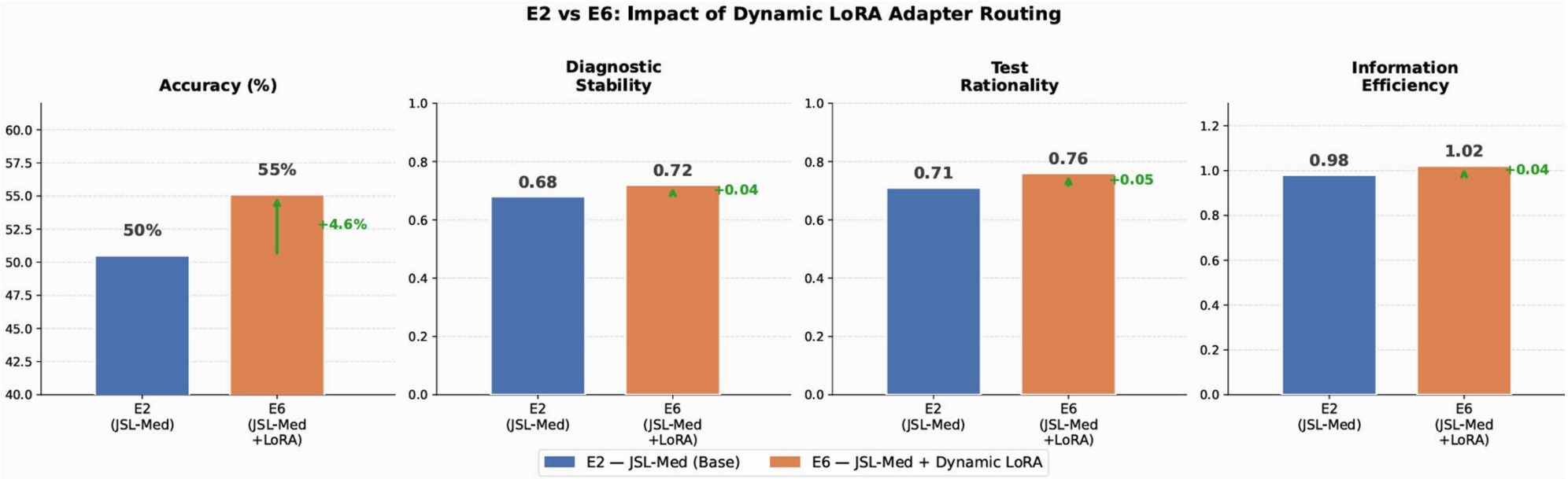
Baseline & Foundational Outcomes

Insights:

- The introduction of the metacognitive reflection node provided enough structural scaffolding to raise E1 accuracy to **45.8%**, actively compensating for the removal of Lexical Leakage.
- Instruction-aligned generalist models (E3) outperform purely medically specialized models (E2) in interactive multi-turn tasks



Dynamic LoRA Results



- **Highest Overall Accuracy (54.7%):** The E6 configuration utilizing Dynamic LoRA routing on JSL-Med outperformed all standalone endpoints.
- **Conclusion:** Decoupling medical deduction from conversational dialogue physically at the matrix-adapter layer resolves the specialisation vs. alignment trade-off entirely.

Discussion: Specialisation vs Alignment

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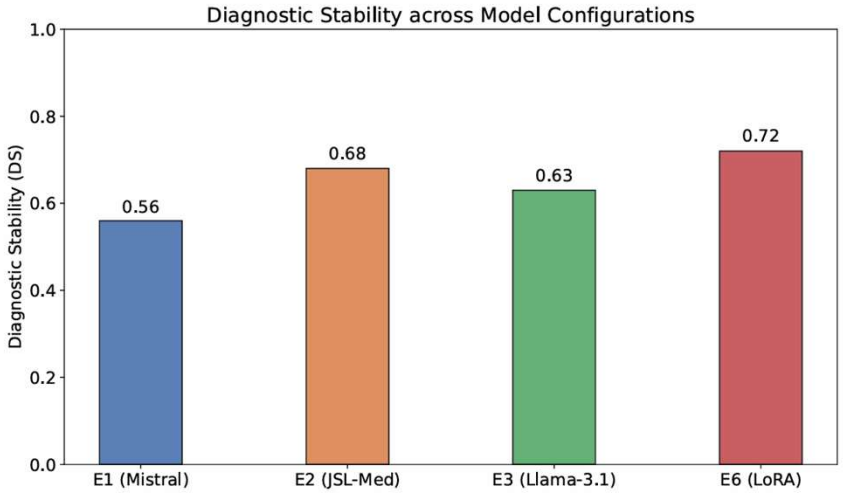
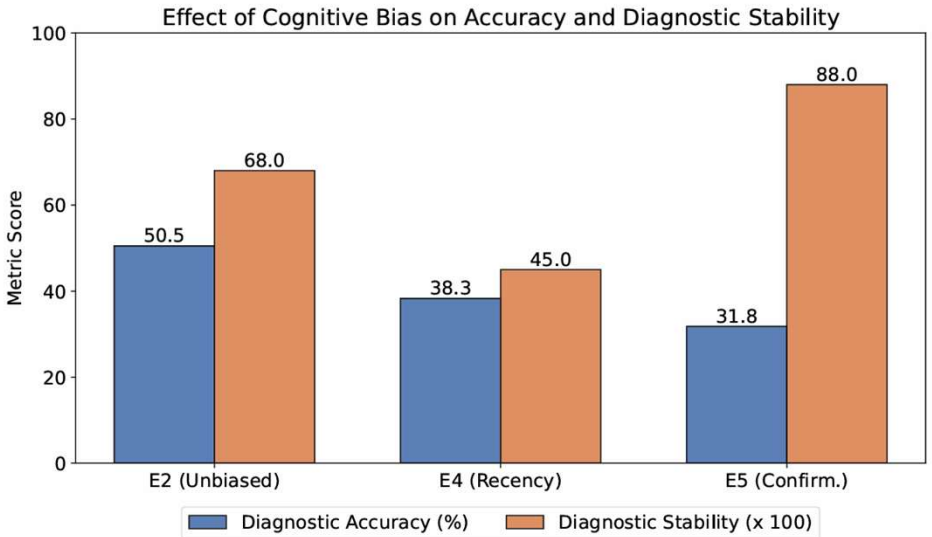
The Safety Threat of Cognitive Biases

Observations from Process-Oriented Measurement:





- **Recency Bias (Accuracy = 38.1%):** Hypothesis thrashing causes Diagnostic Stability to collapse (DS = 0.45).
- **Confirmation Bias (Accuracy = 31.4%):** The lowest accuracy in the study, yet it presents pathologically high stability (DS = 0.88). It rigidly ignores contradictory clinical evidence (premature closure)

Discussion: Specialisation vs Alignment

- **The Trade-off is Avoidable:** E6 proves that the traditional tension between domain-specific fine-tuning and generalized instruction following is an artifact of monolithic model updates, not an intrinsic limitation of Large Language Models.
- **Evaluating Process is Mandatory:** Accuracy alone is dangerous. An model exhibiting high terminal accuracy but generating DS metric signatures indicative of premature hypothesis closure is wildly unsafe for real-world deployment.
- **Open-Source Viability:** Local, quantised LLMs orchestrating across a LangGraph topology represent a scalable, highly secure alternative to proprietary, cloud-hosted API ecosystems.

Conclusion

- **AgentClinic** successfully moves clinical AI evaluation away from static factual recall and towards dynamic, trajectory-aware simulation.
- Implemented heterogeneous multi-engine isolation to resolve the mathematically proven Lexical Leakage vulnerability.
- The introduction of the **Latent Metacognitive Reflection Node** makes LLM reasoning observable and quantifiable.
- While open-source models achieved peaks of **54.7%** diagnostic accuracy (E6 Dynamic LoRA), their severe susceptibility to confirmation and recency bias indicates they are **not yet ready** for direct, unmonitored clinical deployment.

Future Work

The AgentClinic framework serves not merely as an evaluator, but as an advanced foundation for post-training alignment:

- **Direct Preference Optimisation (DPO):** We propose deploying a Process Reward Model. By heavily punishing consultation paths exhibiting low Test Rationality, DPO can actively train the LLM's behavioral policy to prefer systematic, guideline-concordant reasoning sequences over premature guessing.
- **Self-Play AI Feedback:** Eliminating human-annotation dependency. By employing AgentClinic as an automated Reinforcement Learning environment, heterogeneous clinical agents can autonomously generate thousands of synthetic consultation trajectories, using DS and TR process outcomes as reward signals.